SQL THEORY

1. What is SQL?
   SQL stands for Structured Query Language which is used to communicate with the database.

2. What are the types database model?
   Hierachical model (Tree)
   Network model (Graph)
   Document-oriented model (json)
   Object-oriented model (class and objects)
   Relational model (Table)

3. Difference b/w DBMS and RDBMS?

DBMS	RDBMS
Supports file format	Supports table format
Supports single user	Supports Multiple user
No relationship b/w data	Relationship exists b/w data
Doesn't supports meta data , Normalization , ACID properties	supports meta data , Normalization , ACID properties
Low security	High security

4. Difference b/w Primary key and foreign key?

Primary key	Foreign key
Used to uniquely identify the records	Used to build the relationship between the tables
Doesn't allow duplicates and null values	Allows duplicates and null values
Table can have one primary key	Table can have multiple foreign keys
Doesn't depends on Foreign key	Depends on primary key of its parent table

5. Difference b/w Primary key and Unique constraint?

Primary key	Unique
Used to uniquely identify the records	Used to apply uniqueness to the column
Doesn't allow duplicates and null values	Allows null values
Table can have one primary key	Table can have multiple unique constraints

6. Difference b/w DROP and PURGE?

DROP	PURGE
Used to remove the table from database	Used to remove the table from recyclebin
Dropped table can be restored using Flashback	purged table can't be restored
Drop is independent of purge	Purge is dependent on drop

7. Difference b/w TRUNCATE and DELETE?

TRUNCATE	DELETE
Used to remove all the records from the table	Used to remove the specific records from the table
Truncated data can't be recovered	Deleted data can be recovered (only if not commited)
Faster	slower
Its DDL command	Its DML command

8. What is the use of COMMIT, ROLLBACK, LIKE and IS?
    - Commit is used to save the DML operations permanent into the database.
    - Rollback is used to undo the DML operations.
    - Like operator is used for pattern matching.
    - Is operator is used to check the presence of null value.

9. Difference b/w SRF and MRF?

1. SRF (Single Row Function) is known as scalar function.
    - Takes 'n' no_of_rows as input and gives 'n' no_of_rows as output.
2. MRF (Multi Row Function) is known as aggregate/group function.
    - Takes 'n' no_of_groups as input and gives 'n' no_of_rows as output.

10. Difference b/w WHERE and HAVING clause?

where	having
Used to filter the rows	Used to filter the groups
Executes before group by	Executes after group by
Doesn't support MRF	Supports MRF
Independent of group-by	Dependent on group-by

11. What is the order of writing the query & execution of the query?

Writing order	Execution order
1. Select 2. from 3. where 4. group by 5. having 6. order by	1. from 2. where 3. group by 4. having 5. Select 6. order by

12. Difference b/w Sub Query and Joins?

Sub query	joins
Query written inside another query is known as subquery	Fetching the data from multiple tables is known as joins
Mainly used when question has unknowns	Used when data has to be displayed from more than one table
Slower for complex logics	Faster even in complex logic

13. Difference b/w Inner join and Natural join?

Inner join	natural join
Gives only the matched records if joining condition is satisfied	Gives matched records if common column exists between the tables
Supports duplicate columns in the result	Common columns are displayed once at the beginning of the result
Supports tablename.*	Doesn't Supports tablename.*

14. Difference b/w Inner join and Outer join?

Inner join	Outer join
Gives only the matched records if joining condition is satisfied	Gives matched records and unmatched records
Null records are not created	Null records are not created where unmatching is found

15. Difference b/w Inline subquery and CTE?

Inline subquery	CTE
Used in from clause	Used in with clause
Name is optional	CTE Name is mandatory
Not recommended for complex logic	recommended for complex logic
Temp_table can't be reused with in the same query	CTE can be reused with in the same query

16. Difference b/w Rank and Dense rank?

Rank()	Dense_rank()
Provides gap during ranking if ties exits	Doesn't Provides the gap during ranking even if ties exits
Not recommended	recommended

17. What is Normalization and its types?
    Process of organizing the data into database to avoid redundancy and inconsistent dependencies
    - 1NF
    - 2NF
    - 3NF
    - BCNF

18. What is Partial and transitive functional dependencies?
    1. If non-prime attribute is depending on part of the primary key then it is known as Partial functional dependency.
       Can be removed from 2NF
    2. If non-prime attribute is depending on another non-prime attribute then it is known as transitive functional dependency.
       Can be removed from 3NF

19. Difference b/w Left outer join and right outer join?

Left outer join	right outer join?
Gives all the records from left table and only the matcheds records from right table	Gives all the records from right table and only the matcheds records from left table

20. Explain the working of Co-related Sub Query?
    - Inner query executes for each and every record of outer query.

21. Difference b/w Table and View?

table	view
Stores data physically in the database.	Does not store data; just a virtual table based on SQL query.
Can perform INSERT, UPDATE, DELETE operations	Usually used only to read data
Uses storage space.	Does not use extra storage (unless it's a materialized view).
Data is directly stored and managed.	Displays data fetched from one or more tables.

22. Explain ACID properties?

Atomicity: All operations in a transaction are completed or none.
Consistency: Data remains consistent before and after a transaction.
Isolation: Each transaction is independent of others.
Durability: Once committed, changes are permanent even after system failure.

23. Difference b/w Stored Procedure and Function?

Stored Procedure	Function
Can perform DML operations.	Mainly used for calculations, not recommended for DML
May or may not return a value.	Always returns a single value.
Called using EXEC or CALL inside the block	Used in queries like SELECT , WHERE
Supports transaction control.	Doesn't support transaction control (COMMIT, ROLLBACK).

24. Write Definition and types for the given below

1) Block
2) Trigger
3) Function
4) Stored procedure
5) Cursor
6) Index
7) View

1. Block:
   - A group of SQL and PL/SQL statements treated as a single unit.
   - Types: Anonymous block, Named block.
2. Trigger:
   - A PL/SQL procedure that automatically executes when an event (INSERT, UPDATE, DELETE) occurs.
   - Types: BEFORE trigger, AFTER trigger, INSTEAD OF trigger.
3. Function:
   - A PL/SQL block that returns a single value and can be called in queries.
   - Types: Scalar function, Aggregate function.
4. Stored Procedure:
   - A named PL/SQL block that performs tasks and can return zero or multiple values.
   - Used for reusable logic in applications.

5. Cursor:
    - A pointer that helps fetch query results row-by-row in PL/SQL.
    - Types: Implicit Cursor, Explicit Cursor.
6. Index:
    - Improves the speed of data retrieval.
    - Types: B-tree Index, Bitmap Index, Unique Index, Composite Index.
7. View:
    - A virtual table based on SQL query. It simplifies complex queries.
    - Types: Simple View, Complex View, Materialized View.

25. What is the use of Merge Statement?

MERGE is used to combine INSERT, UPDATE, and DELETE in a single statement.

It is helpful when syncing two tables.

26. What is Clustered Index

- A clustered index determines the physical order of data in a table.
- There can only be one clustered index per table because the data rows can be sorted in only one order.
- This index is usually created on the primary key by default.
- Searching becomes faster because the data is stored in the same order as the index.

27. What is Non-Clustered Index

- A non-clustered index stores pointers to the actual data.
- It is separate from the data itself and can be created on any column(s).
- Multiple non-clustered indexes can exist per table.
- They provide faster access to frequently queried columns without changing the actual data order.

28. What are the Advantages of Index

1. Enhances the speed of retrieval operations on a database table.
2. Helps in quickly locating data without scanning every row.
3. Improves performance of SELECT queries.
4. Helps in enforcing uniqueness with unique indexes.

29. What are the Advantages of Cursor
    1. Enables row-by-row processing of records in a result set.
    2. Useful for complex business logic that cannot be performed in set- based SQL.
    3. Can be controlled explicitly using OPEN, FETCH, CLOSE commands.
    4. Can be used in stored procedures and triggers.

30. What are the Advantages of Views
    1. Simplifies complex queries by storing SQL statements as reusable objects.
    2. Enhances security by exposing only necessary columns/data.
    3. Helps in maintaining data consistency and abstraction.
    4. Useful for joining multiple tables and simplifying reporting tasks.

32. When to Avoid Views, Triggers, Cursors, and Indexes

Feature	When to Avoid
Views	For large datasets or complex queries, as they can slow down performance.
Triggers	If they cause delays or hidden logic that is hard to track.
Cursors	For large result sets or when set-based operations can perform better.
Indexes	On small tables (indexes add overhead with minimal benefit). On frequently updated tables (indexes can slow down inserts/updates).

33. What is Data Mining?

Data Mining is the process of discovering patterns, correlations, or trends in large datasets using statistical, machine learning, or database techniques.

- Used for: Customer segmentation, fraud detection, market analysis.
- Involves: Classification, clustering, association rules, regression.

36. What is a Database?

A database is an organized collection of structured data that can be accessed, managed, and updated efficiently using DBMS (Database Management System).

Types:

- Relational (SQL) -- Oracle, MySQL, PostgreSQL , MSSQL
- Non-Relational (NoSQL).-- MongoDB.

37. What is Denormalization?

Denormalization is the process of adding redundancy to a normalized database to improve read performance.

- Reduces the number of joins.
- Suitable for reporting and OLAP systems.
- Trade-off: Improved read speed vs data redundancy.

38. What is a Sequence in SQL?

A Sequence is a database object that generates a series of unique numbers, often used for auto-incrementing primary key columns.

39. D/B COALESCE() and NVL() ?

- COALESCE() when checking multiple values and want the first non-null.
- NVL() when you just want to replace NULL with a default value (Oracle-specific).

40. What is Data Migration?

Data Migration is the process of transferring data between:

- Different systems
- Different formats
- Different storage types

41. OLAP vs OLTP

Feature	OLTP (Online Transaction Processing)	OLAP (Online Analytical Processing)
Purpose	Day-to-day operations	Analysis and decision making
Operations	Insert, Update, Delete	Read-heavy, aggregation
Tables	Normalized	De-normalized
Data Volume	Smaller	Large (historical)
Example	ATM transaction	Sales forecasting dashboard

43. Stages of Transaction in DBMS

Stage	Description
BEGIN / START	Marks the start of a transaction. Changes after this point are part of it.
EXECUTION	SQL operations like INSERT, UPDATE, DELETE are performed.
COMMIT	Makes all the changes permanent in the database.
ROLLBACK	Reverts all changes made during the transaction (if an error occurs).
END	Indicates that the transaction is complete. After COMMIT or ROLLBACK

44. Difference Between DISTINCT and UNIQUE

DISTINCT	**UNIQUE**
Used in a `SELECT` query to remove duplicate rows from the result set.	Used as a constraint to ensure that all values in a column are unique.
Eliminates duplicate rows based on the selected columns in the query result.	Ensures no duplicate values in the specified column in the table.
Does not affect the table structure or data integrity.	Affects the table's data integrity by preventing duplicate values in a column.
Applied only in queries (not in table definitions).	Applied when defining or altering table schema.

45. What is Database Schema

A schema defines how the data is organized and structured within the database